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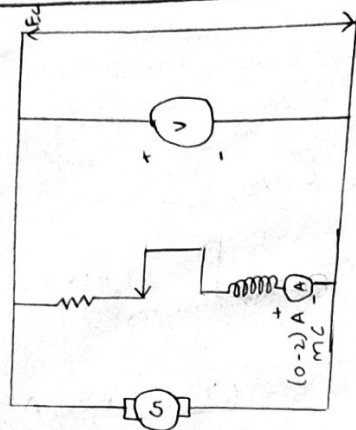
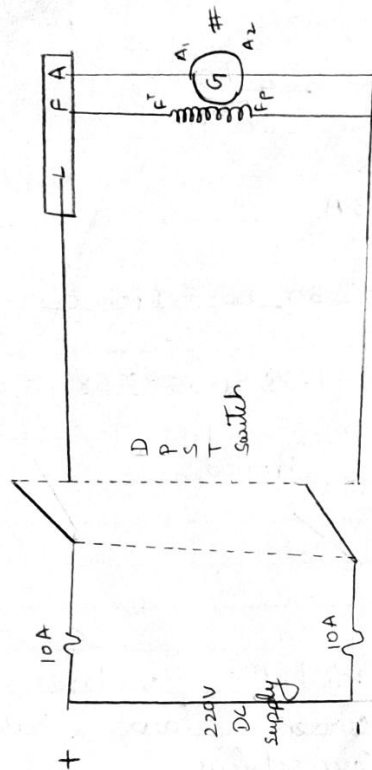
S. No.	Date	Name of the Experiment	Page No.	Marks Awarded	Signature
1	21/1/2026	Verification of Kirchhoff's Voltage Law and Kirchhoff Current Law	99		✓
2	28/1/2026	Open circuit and load Characteristic of DC Shunt generator	99		✓
3	¹¹ 24 /2/2026	Staircase Wiring	99		✓
4	18/02/26	Half wave and Full wave Rectifier	100		✓
5	25/2/26	Zener diode as voltage Regulation	100		✓
6	11/03/26	characteristic of LVDT characteristic variable different transformer.	100		✓

Completed

✓

11/03/2026

OPEN CIRCUIT CHARACTERISTICS



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OPEN CIRCUIT AND LOAD CHARACTERISTICS OF DC SHUNT GENERATOR

AIM

To draw the open circuit and load characteristics of DC shunt generator

APPARATUS REQUIRED

laptop with internet connection

THEORY

DC generator converts mechanical energy into electrical energy

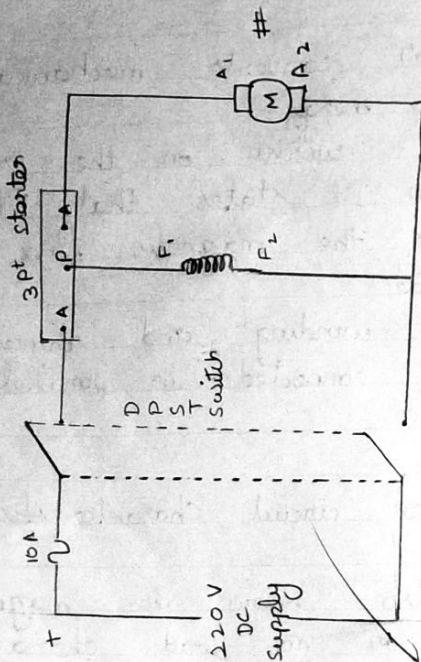
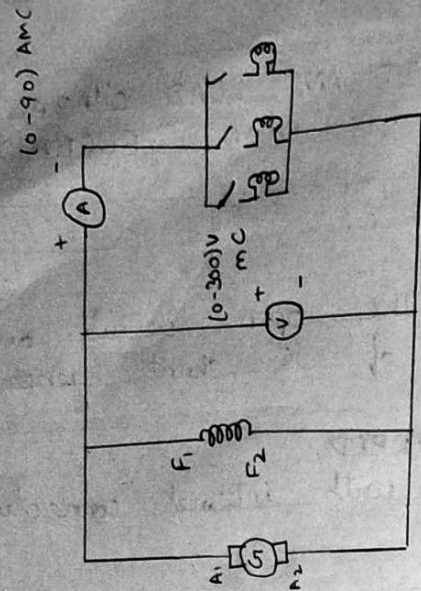
DC generator works on the principle of Faraday's law. It states that "When a conductor cuts the magnetic flux and emf is induced"

The field winding and armature windings are connected in parallel in the shunt generator

Open circuit characteristics

It is also known as magnetisation characteristic or no load characteristic.

Load circuit characteristics



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It is a graph drawn between open circuit voltage (E_o) and field current (I_f)

Load circuit characteristic

It is a graph drawn between load voltage (V_L) and load current (I_L). It is also known as external characteristics

Procedure

Open circuit characteristic

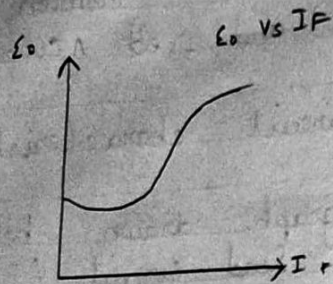
- 1) Connection are made as per the circuit diagram
- 2) switch on the supply and note down the reading (voltage and current)
- 3) By varying the field the Ohm at different values of open circuit diagram voltage and field current are noted
- 4) Plot the graph between open circuit voltage and field current

Load circuit characteristic

- 1) Connections are made as per the circuit diagram
- 2) Switch on the supply and note the no load voltage and current

OPEN CIRCUIT CHARACTERISTICS

MODEL GRAPH



Observation Table

S.NO	voltage	current
1	115	0.16
2	120	0.18
3	126	0.20
4	129	0.21
5	133	0.23
6	135	0.24

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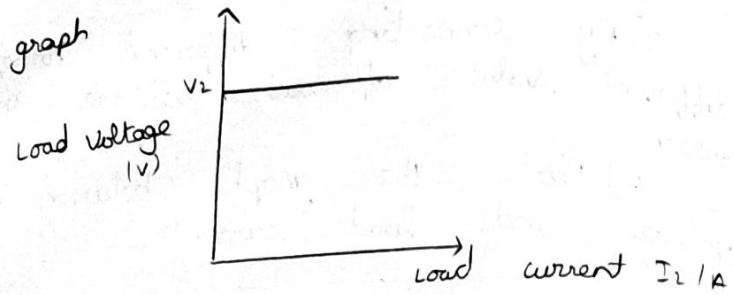
3) By connecting different nodes, different value of load voltage and current

4) Plot the graph between load voltage and load current



LOAD CHARACTERISTIC

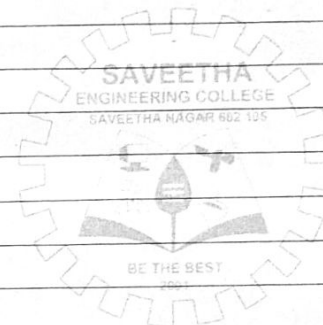
Model graph

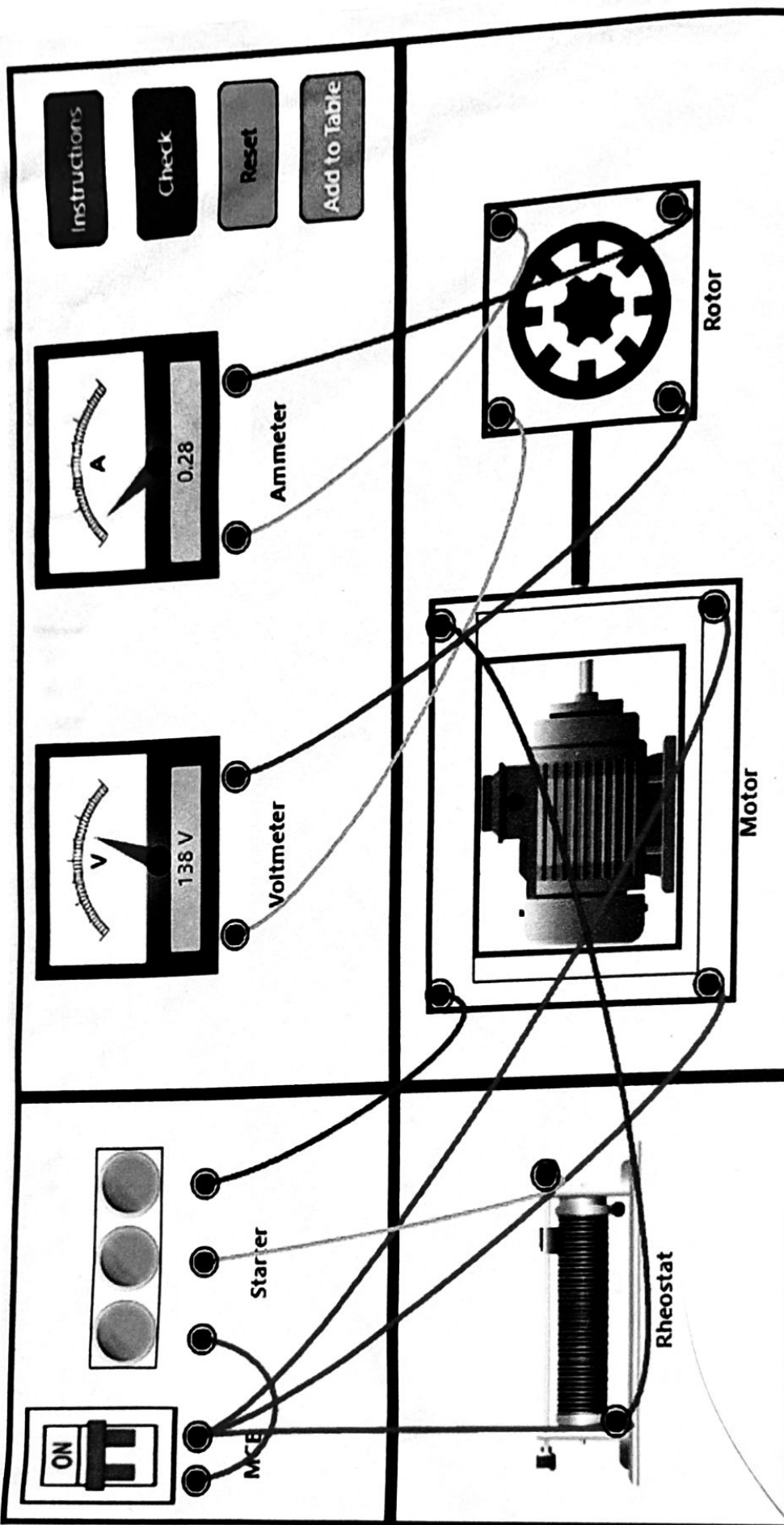


S.NO	current	voltage
1	1.2	220
2	2.8	212
3	3.2	208
4	3.6	205
5	5.5	200
6	7	195

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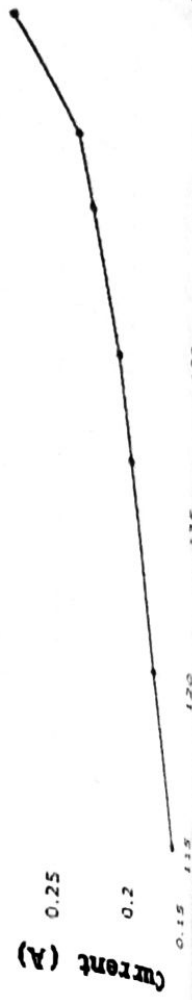


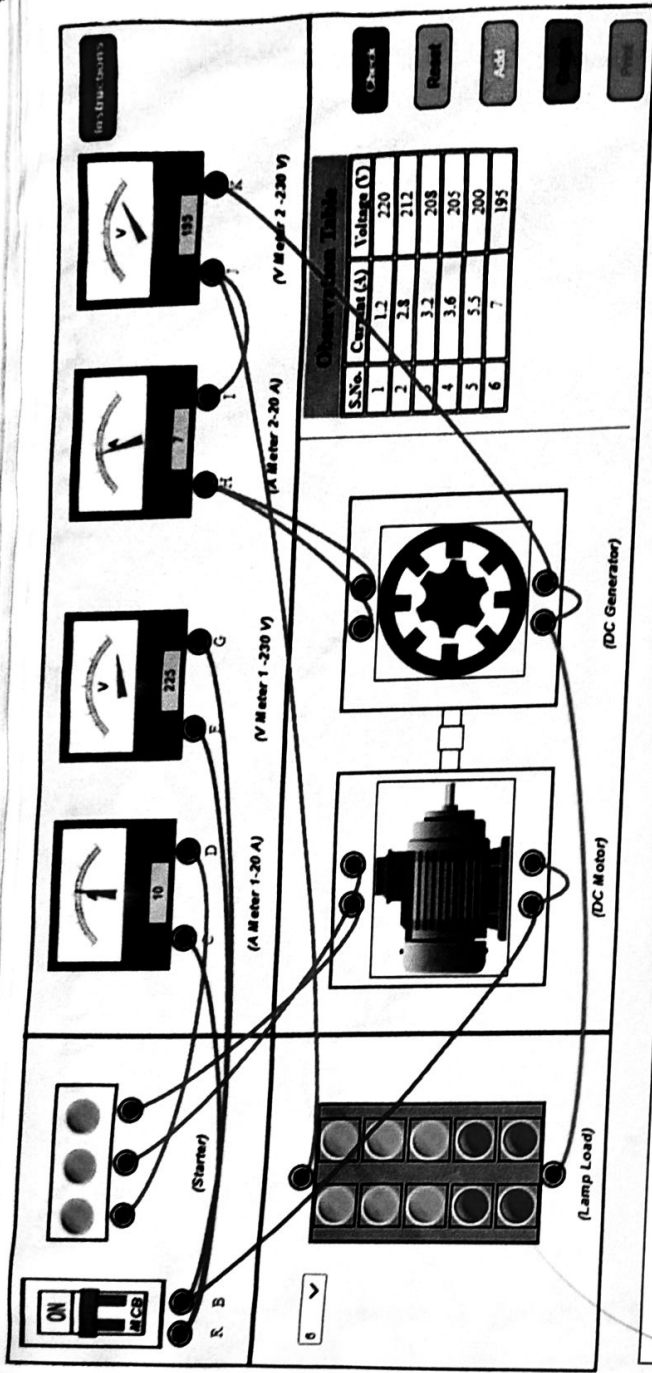


OBSERVATION TABLE

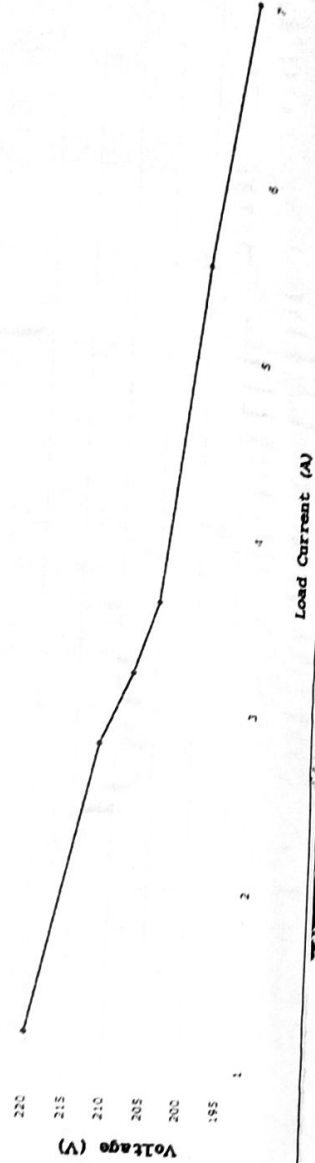
S.No.	Voltage (V)	Current (A)
1	115	0.16
2	120	0.18
3	126	0.20
4	129	0.21
5	133	0.23
6	135	0.24
7	138	0.28

Current (A) vs Voltage (V)





Voltage (V) vs Load Current (A)



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30/01/2016